

Quiz 3

NAME: _____

Q.1. Express $4x^2+8x+5$ in completed square form (i.e. as $a(x+b)^2+c$ where a, b, c are constants).

Q.2. Two children own two-way radios that have a maximum range of 2 miles. One leaves a certain point at 1 : 00 P.M., walking due north at a rate of 4 mi/hr. The other leaves the same point at 1 : 15 P.M., traveling due south at 6 mi/hr. When will they be unable to communicate with one another?

Q.3. Express the complex fraction $\frac{3+5i}{1+i}$ as a single complex number (i.e. as $a + bi$ where a and b are real).

Q.4. Find all real solutions to the equation $2|5x + 2| - 1 = 5$